

**RAJIV GANDHI PROUDYOGIKI
VISHWA VIDYALAYA, BHOPAL (M.P.)
MASTER OF PLANNING (URBAN AND
REGIONAL PLANNING)**

SEMESTER - I

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-111, PLANNING HISTORY AND THEORY

OBJECTIVE

Planning History covers ancient and medieval cities in India and Mughal and colonial built environments in the country. In Planning History course, the following objectives are followed:

- To critically understand economic, political, cultural, social and other forces shaping built environments in history.
- To examine the significance of histories as they inform the present planning of settlements.
- As far as Planning Theory is concerned, the chief goal is not to replace conventional theories with non-conventional ones, but identify under what conditions any set of theoretical propositions hold. Hence, identifying conditions which determine development and planning outcomes is crucial. One assumption that underlies this pedagogical approach is that it encourages unconventional thinking, and reflection in action—and not planning prescriptions based on pre-conceived ideas. Following are the course objectives of Planning Theory:
- To learn to have a general, and substantive command of the spectrum of past and present perspectives regarding planning theory.
- To learn to write clearly and concisely about abstract planning concepts.

CONTENTS

- **Unit 1: Ancient and Medieval Cities in India**
Urbanization in the Medieval period including a study of cities like Varanasi, Mohenjo-Daro, Harappa, Dholavira and Lothal; Chola dynasty sponsored famous towns like Thanjavur, Nagapattinam, Kanchipuram and Madurai; Why India could not become urbanized during the medieval period? Indian knowledge systems in town planning, 'nagara Vaastu Shastra'.
- **Unit 2: Recent History of Indian Cities**
Mughals and Vijayanagara Kingdoms built grand cities like Shahjahanabad in the north and Hampi – the capital of Vijayanagara Kingdom in the south; Mughal towns and cities; City and regional planning during the British Raj involving cantonments, model towns, capital towns like New Delhi; Regional planning efforts involving development of port cities such as Bombay, Madras, Calcutta, etc.; Construction of railways, and road networks; The Portuguese port towns of Cochin, Cannanore, Goa, Daman and Diu. Contributions of scholars like Sir Patrick Geddes to Indian planning; Governance and planning legislation during the British period include a study of improvement trusts, municipalities, etc., contributions of Patrick Geddes, Le Corbusier, Koneisberger, etc. in modern Indian city planning.
- **Unit 3: Urban Theories**
Concentric Zone Theory, Sector Theory, and Multiple Nuclei Theory with a focus on the contributions of Robert E. Park, Louis Wirth, etc.; Land Use and Land Value Theory of William Alonso; From the world city to the global city.
- **Unit 4: Theories of Planning**
Rational Planning Model; Advocacy Planning Model; Political Economy Model; Equity Planning Model; Radical Planning model; Collaborative Planning Theory.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- To demonstrate appreciation and knowledge of histories of planning.
- To explain and analyse abstract theoretical formulations.

REFERENCES

- Allmendinger, P. (2017) *Planning Theory*, Macmillan, London. Third Edition.
- Buch, M. N. (1987) *Planning the Indian City*, Vikas Publishing, New Delhi.
- Fainstein, S.S. and DeFilippis, J. (Eds.) (2016) *Readings in Planning Theory*, Wiley, London. Fourth Edition.
- Friedmann, J. (1987) *Planning in the Public Domain*, Princeton University Press, Princeton. Second Edition.
- Gooptu, N. (2001) *The Politics of the Urban Poor in Early Twentieth - Century India*, Cambridge University Press, Cambridge.
- Healey, P. (2010) *Making Better Places: The planning project in the twenty-first century*, Palgrave Macmillan, London.
- King, A. (1976) *Colonial Urban Development: Culture, Social Power, and Environment*, Routledge and Kegan Paul, New York.
- Kumar, A., Sanjeev, V., and Prakash, P. (2020) *City Planning in India, 1947-2017*, Routledge, New York.

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-112, SOCIO-ECONOMIC BASIS FOR PLANNING

OBJECTIVE

- To provide exposure to concepts, theory and issues relating to socio-economic aspects towards planning of settlements. Also provide understanding of the society and the economy of the nation and its importance in spatial planning.

CONTENTS

- **Unit 1: Introduction to Sociology**
Definition and scope of sociology; Concepts of sociology-society, social systems, social structure, institution and organization; Concept of space and people; Sociology and planning.
- **Unit 2: Social Groups, Social Issues, Rural and Urban Sociology**
Contemporary sociological theories; Social structure and social change; stratification and social inequality; Introduction to agrarian, industrial and modern society and spatial formation; Linking social structure and physical structure of village and urban settlements; marginality, vulnerability, social inclusion and exclusion; Inequality and equity; Children youth and gender centred planning.
- **Unit 3: Applied Economics – Goods and Services**
Definition of economics – terms used in economics related to urban and regional planning (URP); central problems of economics; Basics of micro and macroeconomics; use of economics in planning; Definition of need, demand and supply; Law of demand and supply, types of demand; Theory of demand and utility; Elasticity of demand and supply – its use in planning; Types of economics and their application in spatial planning.
- **Unit 4: Land Economics**
Economic concepts of land, objectives and scope of land economics, relevance of spatial planning; Economic principles of land uses; Economic rent, land use and land values, market mechanism and land use pattern.
- **Unit 5: Economics of Location and Planning**
Analysis of location of specific uses like residential, industrial, commercial and institutional in the light of location theories in intra-regional context; Techniques of cost-benefit analysis of urban development programme.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- Gain a comprehensive understanding of the concepts, theories, and issues related to socio-economic aspects relevant to planning settlements.
- To evaluate the economic implications of location choices for residential, industrial, commercial, and institutional developments.
- To utilize cost-benefit analysis techniques to assess the feasibility and impact of urban development programs.

REFERENCES

- Benjamin S (2008), *Occupancy Urbanism: Radicalizing Politics and Economy beyond Policy and Programs*, *International Journal of Urban and Regional Research*, Vol. 32:3 September, 719-729.
- Brenner N. and Theodore N (2002), *Cities and Geographies of "Actually Existing Neoliberalism"*, *Antipode*, Vol. 34, Issue 3, 349-379
- De Souza M (2010), *Which Right to Which City? In Defence of Political-Strategic Clarity*, *Interface*, Vol. 2(1), May, 315-333.
- Jan L., Christopher M. (2012), *The Urban Sociology Reader*, Routledge, London.

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-113, PLANNING TECHNIQUES

OBJECTIVE

- This course intends to impart knowledge about preparation of maps and undertake data collection and analysis for plan preparation.

CONTENTS

- **Unit 1: Scales and Preparation of Maps**
Maps as a representation of reality, Elements of Maps; Graphical, linear and areal scales, Notations involving basic discipline of maps; Measurement of areas; Learning to prepare base maps; Contents of base maps at various scales; Choice of appropriate scales for region and settlement level plans, town development plans, zonal development plans, layout plans.
- **Unit 2: Data for Planning and Socio-Economic Surveys**
Data requirements for urban and regional planning; Sources of primary and secondary data; Quantitative and Qualitative methods of data collection, Validity and reliability of data, Questionnaire design, measurement scales and their applications, sampling techniques, types of socioeconomic surveys; Self-surveys, interviews, mailed questionnaires and observer participation, focus groups etc.
- **Unit 3: Physical Surveys**
Techniques of conducting surveys for land use, building use, density, structural condition of buildings, heights of buildings, land utilization and physical features of land; Data requirements for various types of regional plans; Techniques for conducting regional surveys.
- **Unit 4: Analysis and Presentation of Data**
Land suitability analysis, Land use classification, coding and analysis; residential and non-residential density patterns and their analyses; population and economic analysis. Tabulation of data, graphical presentation of data; Preparing pie diagrams, histograms, bar charts, normal, semi-log and double log graphs and their uses; colour, black and white presentation techniques; Understanding the discipline of illustrations and tables. Colour, black and white presentation techniques; Basic discipline of presenting illustrations; Presentation of spatial data, analysis and proposals., Use of AI and ML, Big data, etc.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- To create base maps and present planning information on maps.
- To analyse the types of data required for planning and methods of data collection.
- To demonstrate an understanding about data analysis and to be able to examine data for understanding the existing situation in a settlement.

REFERENCES

- Bracken, I. (1999) *Urban Planning Methods: Research and Policy Analysis*, Methuen, London.
- Berke, P.R. and Goodschalk, D.R., Kaiser, E.J and Rodriguez, D.A (2006) *Urban Land Use Planning*, University of Illinois Press, U.S.A. Fifth Edition.
- Wang X., Rainer A. and Hofe, V. (2007) *Research Methods in Urban and Regional Planning*, Springer, Berlin.
- Monmonier, M. (1996) *How to Lie with Maps*, University of Chicago Press, Chicago.

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-114, INFRASTRUCTURE PLANNING

OBJECTIVE

- To provide basic knowledge about physical and social infrastructure and requirements of various infrastructure for a settlement.
- To study planning and policy issues for urban and regional infrastructure.

CONTENTS

- **Unit 1: Water Supply System**

Water supply systems and networks, and network mapping; Water sources, quality and quantity requirements, and water requirement for various land uses; Factors affecting water demand; Storage facilities and distribution systems; Rain water harvesting systems and locational criteria, implications on land use and density of water harvesting system; Innovative Methods and successful urban water supply system practices; Water programmes and policies. Traditional and Modern Water Conservation practices., water security, water recycling.

- **Unit 2: Sanitation and Sewerage System and Storm Water Drainage**

General considerations and principle of sanitation and sewerage systems; Sewage disposal and treatment methods; Characteristics of waste water, industrial pollutants and their affects; Open defecation; Manual scavenging; Innovative approaches of sewage disposal in urban areas and low cost appropriate technologies for sanitation; Strom water drainage networks, and network mapping; Estimations of sewer generation and network requirements; Elements of Solid Waste Management, Classification and Characteristics of Solid Wastes; Methods for Solid waste Collection, Storage, transportation and disposal; Processing and Treatment of Solid Wastes; Land Filling methods of Solid Waste Management. Organic treatment of waste, e-waste management. Urban flood control, planning of storm water drainage.

- **Unit 3: Social Infrastructure**

Social infrastructure typologies; Planning norms and space standards for education, health, recreation and socio-cultural facilities; Amenities for urban and rural settlements; Significance of education and health infrastructure in planning; Locating education and health facilities; Understanding scalogram and other techniques.

- **Unit 4: Policies, Programmes and Projects**

Understanding prevalent policies, projects and missions, for example, JnNURM, AMRUT, HRIDAY, Smart Cities Mission, etc.; Norms and standards for different types of infrastructure; Nature and content of infrastructure in development plans at different geographical levels; Making assessment of infrastructure requirements in plans.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- To identify the problems and issues related to infrastructure provision in a settlement.
- To appraise planning guidelines to tackle these problems, and extent of infrastructure required for the future planning of settlements.

- To design and make recommendations for meeting the future needs infrastructure for a human settlement.

REFERENCES

- Hudson, W.R., Hass, R.C.G. Uddin, W. (1997) *Infrastructure Management*, McGraw Hill, London.
- Gifford, J.W. Uzarski, D.R. and McNeil, S. (1993) *Infrastructure Planning and Management*, American Society of Civil Engineers, Reston, VA.
- Goodman, A. and Hartak, M. (2000) *Infrastructure Planning Handbook*, ASCE Press, Reston, VA.
- Parkin, J. and Sharma, D. (1999) *Infrastructure Planning*, Thomas Jelford Publishing, London.

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-115, HOUSING AND ENVIRONMENTAL PLANNING

OBJECTIVE

- The first part of this course provides an understanding about the nature of housing problems, and various programmes and policies initiated to deal with these problems in Indian cities and villages.
- The second part focuses on the significance of physical environment and its various parameters as they relate to plan preparation and implementation processes.

CONTENTS

- **Unit 1: Problem of Housing**
Significance of housing in economic and social development; Basic Terms in housing, understanding concepts of Housing need, demand, supply and shortage; Housing data analysis from Census; Housing typologies; Housing development process; Nature of housing problem in urban and rural India, Public health issues in housing; Housing for the poor, houselessness in India, old housing,
- **Unit 2 Housing Policy and Finance**
Evolution of housing policy in India, Role of the government and private sector in housing; housing programmes for the poor and night shelters; Examples of housing projects, Introduction to housing finance; Affordability and Cost of housing; Role of Financing organisation. Critical Review of Current housing programmes, National Housing Policy, Various State Housing Policies and Affordable Housing Policies, PMAY Housing for All.
- **Unit 3: Ecosystem, Urban Ecosystem and Quantitative Ecology**
Eco-system and their relevance to environment, Impact of advanced agriculture-methods, urbanization and industrialization on nature; Pollution, types, sources, remedies; Urban ecosystem approach, evolution and significance; Introduction to quantitative ecology, Identification of ecological parameters for planning at different levels, Site planning, Settlement planning, regional planning. Data needs, formats for data collection Types of analysis required to evolve ecological parameters; Environmental impact assessment (EIA), Methods and their appraisal. preparation and analysis of resource inventories and resource matrices.
- **Unit 4: Resource Planning and Management**
Finiteness of resources, examples of transfer from one resource to another in history in different parts of world; Development, utilization and conservation of resources, resource planning, integrated resource planning approach; Resource regions, their problems and potentials; Resource management, traditional and contemporary approaches. Resource development in India, some selected areas (energy, water, manpower, etc.); and Government's 8 missions under the National Action Plan on Climate Change.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- To investigate and analyse the nature of housing problem in India.
- To examine the housing policy and programmes in India.
- To analyse the eco-system and resources and their importance for planning.
- To appraise the environment parameters for consideration while planning.

REFERENCES

- Cedric, P. (1990) *Housing and Urbanisation: A Study of India*, Sage, London.
- Hardoy, J.E. and Satterthwaite, D (1989) *Squatter Citizen: Life in the Urban Third World*, Routledge, London.
- Kohli, V.K. (2007) *Housing Finance Agencies in India*, Deep and Deep, New Delhi.
- Jenkins, P., Smith, H. and Wang, Y.P. (2007) *Planning and Housing in the Rapidly Urbanizing World*, Routledge, London.
- McHarg, I. (2008) *Design with Nature*, Wiley, London. 25th Edition.
- Ndubisi, F.O. (ed.) (2014) *The Ecological Design and Planning Reader*, Island Press, Washington, D.C.
- Wallace-Wells, D. (2019) *The Uninhabitable Earth*, Crown Publishing, U.S.A.
- Wachter, S.M. and Birch, E.L. (2008) *Growing Greener Cities: Urban Sustainability in the Twenty First Century*, University of Pennsylvania Press, Philadelphia, PA.
- Whitehead, M. (2014) *Environmental Transformation: A Geography of the Anthropocene*, Routledge, New York.

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-116, AREA STUDIO

OBJECTIVE

- Area planning studio intends to develop an ability in students to apply the concepts learnt in theory subjects to day-to-day planning practice. Through field visit, the students would understand comprehensive, interrelatedness and long-term nature of planning in a settlement.

CONTENTS

- This course provides basic introduction to GIS and Statistics to be applied to relevant exercises. Developing an appreciation of components of a settlement and develop an ability to read a city or small settlement.
- Second part of the studio will require the students to prepare a development plan for a neighbourhood or village for rural and urban areas based on a field visit. As planning endeavour is a team work, it is expected in this studio, the students make a beginning to learn to work in collaborative environments.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- To develop data collection and analytical skills through field surveys and analysis for the purposes of plan preparation.
- To apply theoretical concepts to the real-world situations.
- To develop sensitivity to group dynamics and working in teams.

RAJIV GANDHI PROUDYOGIKI VISHWA VIDYALAYA, BHOPAL (M.P.) MASTER OF PLANNING (URBAN AND REGIONAL PLANNING)

SEMESTER – I

URP-117, GIS FOR PLANNING

OBJECTIVE

- To equip with the concepts of geo-informatics and computing skills in the relevant software, associated scientific tools and their relevance and applicability in urban and regional planning.

CONTENTS

- **Unit 1: Introduction to Geo-informatics**
Definitions – Geoinformatics, Remote sensing, Geographic Information System (GIS), Spatial Data Infrastructure; the concept of earth surface projections and geoids; limitations of DBMS, engineering drawings and CADD packages – the need for GIS, Spatial and non-spatial data, raster and vector data, spatial thematic models.
- **Unit 2: Remote sensing**
Remote Sensing (RS) – Types of RS: passive and active, electromagnetic spectrum; platforms – space, air and ground; data capturing concepts (multi return concepts), digital images – satellite and aerial photography, resolutions, geo-referencing, projection, image processing-enhancement, rectification, transformation; image classification and analysis, digital elevation models, photogrammetric methods-integration, application, stereo data analysis, stereoscopy and triangulation.
- **Unit 3: Geographic Information Systems**
Components of a GIS; spatial and attribute data – input and output; spatial data entry – data structure for GIS, vector data structures; Coordinate systems; Geodetic data – point positioning, problems, measurements, spatial analysis using lab modules, etc.; Relevant Spatial analysis software, Data creation and query; Map preparation – Geo-referencing, digitization, scales, layers, layout, topology creation, spatial data analysis – buffer, overlay, 3D analysis and modelling; Emerging and advanced technology – web-enabled GIS, GPS tracking and monitoring, model builder, transparency through GIS, community participation through GIS, monitoring and management, mobile geo-spatial data collection, aerial mobile mapping, emergency response planning.
- **Unit 4: Concepts of Spatial Data Infrastructure (SDI)**
Framework of geo-spatial data, interactively connected users and tools, Agreements on geo-spatial standards, Policies to facilitate access to geo-spatial data by users, Institutional arrangements, Use of SDI to communicate spatial data, issues, guidance and services for urban and regional planning.
- **Unit 5: Applications in Urban and Regional Planning**
Preparation of base map, land use maps, utility and infrastructure maps, area delineation, cadastral maps, etc.; Area delineation, inventory preparation of classes; Condition assessment of specific areas, Quantitative measurement of landscape surfaces; Vulnerability mapping and Monitoring.

COURSE OUTCOMES

Upon the completion of this course, the students would be able:

- Gain an understanding of fundamental concepts in geo-informatics, including Geographic Information Systems (GIS), Remote Sensing (RS), and Spatial Data Infrastructure (SDI).
- Develop proficiency in utilizing scientific tools and relevant software for data processing, analysis, and visualization in urban and regional planning contexts.
- Explore the components and functionalities of GIS, including spatial data entry, manipulation, analysis, and map preparation, for effective spatial decision-making in planning.
- Apply geo-informatics tools and techniques to various aspects of urban and regional planning, such as base map preparation, land use mapping, infrastructure inventory, and vulnerability assessment, to support informed planning decisions and policy formulation.

REFERENCES

- Anil K. Jamwal (2008), *Remote Sensing and GIS*, Jnanada Prakashan, Delhi.
- Cambell J.B. (2002), *Introduction to Remote Sensing*, Taylor & Francis, London.
- Jan Van Sickle (2010), *Basic GIS Coordinates*, Second Edition, CRC Press; 2 Edition, NY.
- Richards J.A. and Xia X. (2006), *Remote Sensing Digital Image Analysis: An Introduction*, Birkhauser, London.